

EE 671: VLSI Design

ASSIGNMENT 2



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1.1 Layout

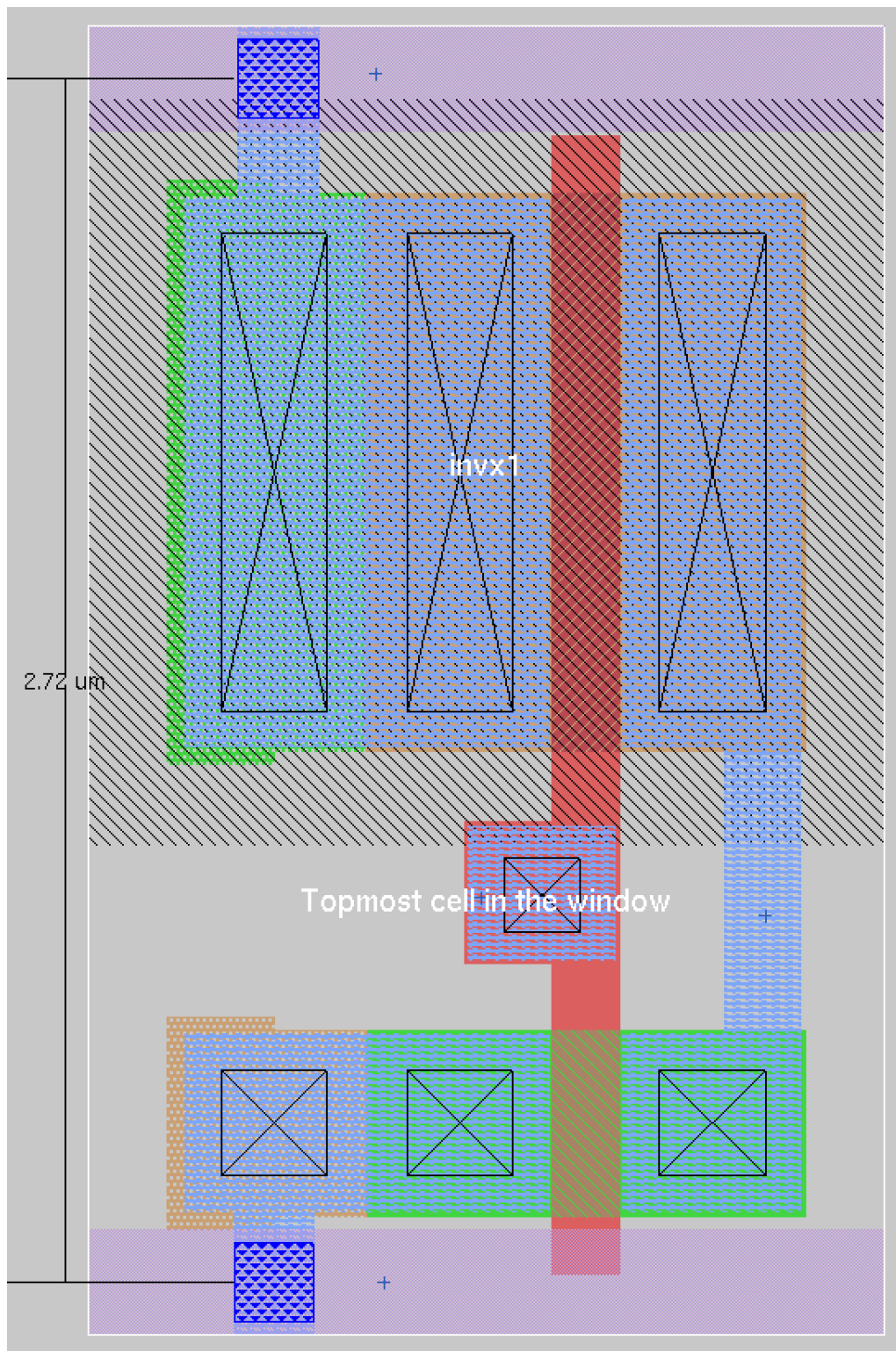


Figure 1: INVX1 layout.

1.2 DRC



Figure 2: INVX1 DRC check.

1.3 LVS

```
Contents of circuit 1: Circuit: 'invx1'
Circuit invx1 contains 2 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances: 1
  Class: sky130_fd_pr__pfet_01v8 instances: 1
Circuit contains 4 nets.
Contents of circuit 2: Circuit: 'not1'
Circuit not1 contains 2 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances: 1
  Class: sky130_fd_pr__pfet_01v8 instances: 1
Circuit contains 4 nets.

Circuit 1 contains 2 devices, Circuit 2 contains 2 devices.
Circuit 1 contains 4 nets, Circuit 2 contains 4 nets.

Final result:
Circuits match uniquely.
.
Logging to file "comp.out" disabled
LVS Done.
```

Figure 3: INVX1 LVS check.

1.4 Dynamic Characteristics

Table 1: Dynamic characteristics of INVX1.

Characteristic	Value
Rise time, t_r (ps)	23.372
Fall time, t_f (ps)	23.878
Propagation delay, t_p (ps)	25.609

2 INVX2

2.1 Layout

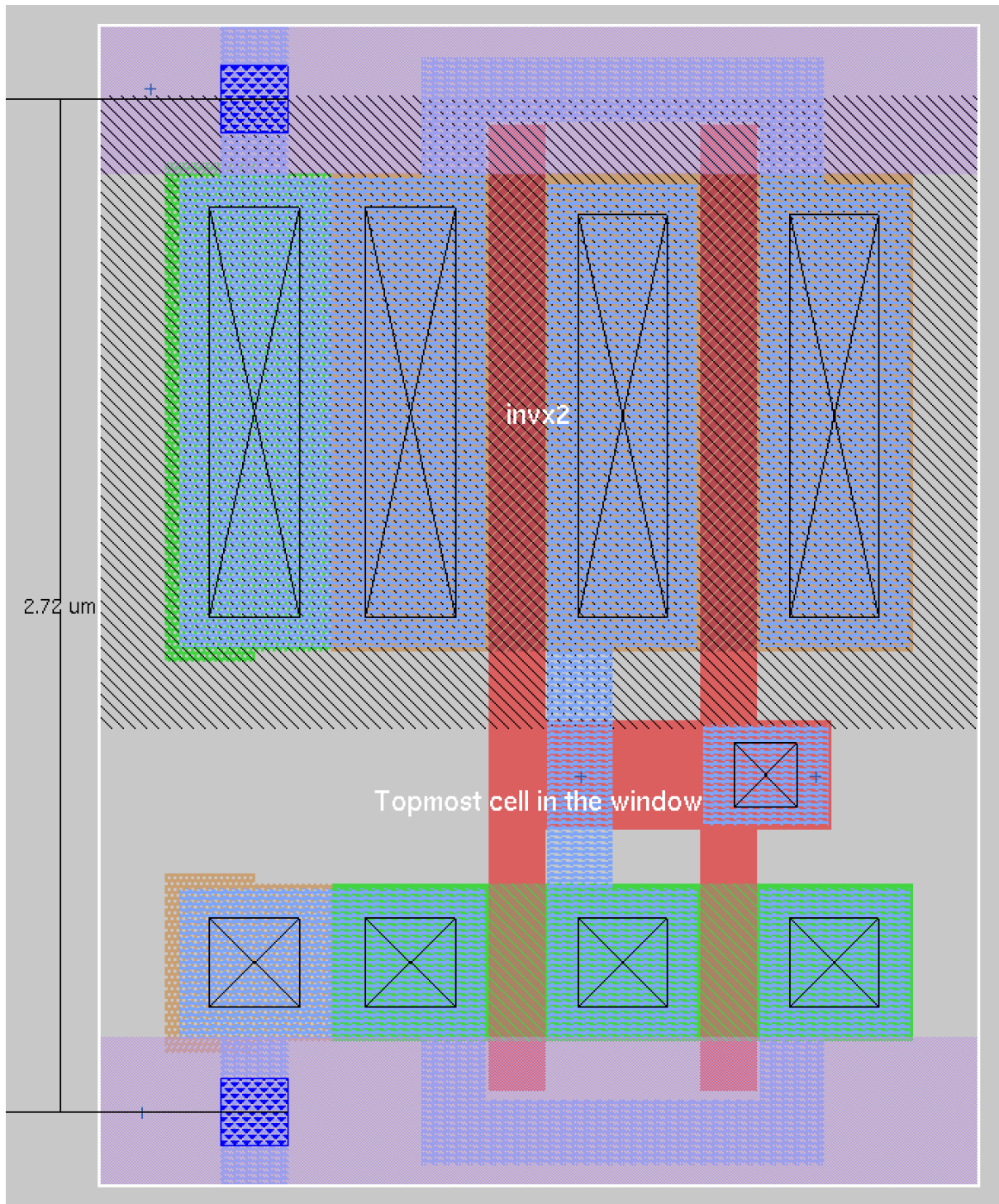


Figure 4: INVX2 layout.

2.2 DRC



Figure 5: INVX2 DRC check.

2.3 LVS

```
Contents of circuit 1: Circuit: 'invx2'
Circuit invx2 contains 2 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances: 1
  Class: sky130_fd_pr__pfet_01v8 instances: 1
Circuit contains 4 nets.
Contents of circuit 2: Circuit: 'not2'
Circuit not2 contains 2 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances: 1
  Class: sky130_fd_pr__pfet_01v8 instances: 1
Circuit contains 4 nets.

Circuit 1 contains 2 devices, Circuit 2 contains 2 devices.
Circuit 1 contains 4 nets, Circuit 2 contains 4 nets.

Final result:
Circuits match uniquely.
.
Logging to file "comp.out" disabled
LVS Done.
```

Figure 6: INVX2 LVS check.

2.4 Dynamic Characteristics

Table 2: Dynamic characteristics of INVX2.

Characteristic	Value
Rise time, t_r (ps)	13.012
Fall time, t_f (ps)	13.453
Propagation delay, t_p (ps)	18.096

3 Remarks

The **INVX1** cell was designed following the tutorial video. The **INVX2** cell was implemented by increasing the number of transistor fingers to enhance the effective drive strength. In Assignment 1, the width was set to 2.57, but here it was slightly modified to 2.56 to make the design realizable in Magic, without significantly affecting the calculated values.

The small deviations in timing parameters (a few picoseconds) compared to Assignment 1 are due to increased active areas, which were necessary to eliminate DRC violations. For the dynamic characterization, the updated inverter subcircuits generated from Magic were used directly in transient simulations. Overall, the results confirm that the layouts are both **DRC and LVS clean** and achieve the expected functional behavior.